CUMC 2025 Student Abstracts

CUMC Organizing Committee

Mathematics Faculty

University of Waterloo

Canada

June 22, 2025

	Algebra, Geometry and Topology	Analysis, Probability, and Statistics	Applied Mathematics, Mathematical Physics	Combinatorics, Graph Theory, and Logic
Tuesday Morning (Rooms)	MC 4040	MC 4060	MC 4063	MC 4041
9:00 am	Matthias Hannesson: Multiplicative Groups Avoiding a Fixed Group	Carol Wu: Identification of Interventional Distributions		
9:30 am	Jason Fang: Double Centralizer Theorem for Central Simple Algebra	Mark Ewert: LEAP: A Stochastic Microsimulation Model for Asthma Outcomes and Policy in Canada		
10:00 am	Jester Cabading: Lie Theory and the Connection Between Analysis, Algebra, and Geometry	Kai Boisvert: Generalized Symplectic Vector Spaces and Operads		Yuewen Luo: Enumerating Permutations in Even and Odd Symmetric Groups
10:30 am		Sushrut Tadwalkar: Unpacking The Kakeya Conjecture: Problems and Approaches	Mingyue Guo (Gloria): A look at AIM with new applications	

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Tuesday Afternoon (Rooms)	MC 2066	MC 4040	MC 5479	MC 5417
2:30 pm			Aaron Wei: Kernels, Hilbert Spaces and Hypothesis Testing	Jiwei Peng: Spectral Clustering and Image Segmentation
3:00 pm	Taha Hedayat: Post-Quantum Cryptography: The Spine of the Supersingular Ell-Isogeny Graph	Leo Guan: The Martingale Convergence Theorem - A Lesson on Gambling		
3:30 pm	Sara Nayar: An introduction to Gelfand-Kirilov dimension		Mark Choi: Spectral problems related to solitary waves of the nonlinear Schrodinger equation and their stability	
4:00 pm	Frédéric-Alexandre Lacasse: Rescuing polynomials: a differential topology perspective		Manuela Salerno and Niv Karo: 3D Printing Manipulatives in Mathematical Education	
Wednesday Morning (Rooms)	MC 4040	MC 4060	MC 4063	MC 4042

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Table 0: (Continued)

9:00 am		Madeline Forbes: The Polynomial Method and the Finite Field Kakeya Problem		Stephanie Penner: Automating Enumeration: Chord Diagrams from Combinatorics to Code
9:30 am	Alexander Phardis: Trefoil Divisibility: When is a Knot Divisible by the Trefoil?	Nicholas Rees: Glasses for the Mind's Eye		
10:00 am	Tsepella Goff-Martin: Colouring Knots with S4		Manuel (Zicheng) Wang: Optimization and Monotone Operator Theory	
10:30 am	Amos Zhang: Coloring Pretzel Knots Under The Symmetric Group Of Degree 4		Edwin Huras: Cells Navigating in Complex Environments: Models and Simulations	
Wednesday Afternoon	MC 4042	MC 4059	MC 5417	MC 5497
2:30 pm	Axel Liu-Bai: The isoperimetric problem in the plane			Michael Alexander: Reservoir Computing: It Just Works!
3:00 pm		Matthew Bull-Weizel: The High-Low Method in Incidence Geometry	Chris Richard: Fractal Geometries and Architecture	Kai Choi: Alice in Quadraticspanningforestidentityspace

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Table 0: (Continued)

3:30 pm		Cuiwen (Frank) Zhu: Circle method on Waring Problem and Hasse Principle		Hanyang Xiao: Perfect State Transfer in Symmetric Square of the n-cube
4:00 pm	Aria Duggan: Perfect Numbers		Taha Hedayat: Can AI Teach Proof? Student experiences with a custom chatbot in an introductory Algebra course	
Thursday Morning	MC 4040	MC 4060	MC 4063	MC 4041
9:00 am	Ena Bahk-Pi: The Classification of Finite Real Reflection Groups			Emma deBlieck: Kemeny's Constant and Braess' Paradox for Graphs
9:30 am	Brendan Matthews: Geometric Tomography of Convex Polygons using Piecewise Analytic Hedge- hogs	Chunxin Zheng: Solving Integer Programming Problems with Quantum		Edmond Yu: Large numbers and proof theory
10:00 am	Alex Pawelko: How Special is Curvature?			Ewan MacDuff: Eternal Domination of Graphs
Friday Morning	MC 4020		MC 4042	MC 4063

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Table 0: (Continued)

9:00 am			Maria Chzhen: Mathematical Modeling and Reinforcement Learning for Strategic Ocean Plastic Retrieval	Mithuna Threz and Arul Milton: 2-dimensional partial orders and ultrahomogeneity
9:30 am	Nathan Acheampong: The Undecidability of the Word Problem for Groups			Jonny Aloni: Hermitian adjacency matrix
10:00 am			Carter Clifton: Understanding Opioid Mortality during COVID-19 - An Individual-Based Model	Amir Dadpour: Introduction to Interdiction Optimization

Algebra, Geometry, and Topology (MC 4020)

Multiplicative Groups Avoiding a Fixed Group

Matthias Hannesson

We know that any finite abelian group G appears as a subgroup of infinitely many multiplicative groups $\mathbb{Z}_n^\times$. It seems to be less well appreciated that G appears as a subgroup of almost all such groups. In this talk we will explain this result and present an asymptotic formula for the number of exceptional n where $\mathbb{Z}_n^\times$ fails to contain G as a subgroup. The work presented in this talk was done in joint with Dr Greg Martin of UBC.

How Special is Curvature?

Alex Pawelko

Why isn't a sphere flat? How can we prove this? One way is through the failure of parallel transport to be trivial: if you take a tangent vector to the north pole of the sphere, then move it along a closed loop while keeping it parallel without turning, the vector might point in a different direction when one returns to the north pole. This phenomenon is called holonomy, and is a manifestation of the sphere being curved. In this talk, we will give an introduction to the subfield of differential geometry called special holonomy, which studies how certain spaces have "special" curvature phenomenon, closely related to questions of when these spaces have objects obeying special PDEs.

Trefoil Divisibility: When is a Knot Divisible by the Trefoil?

Alexandros Phardis

The trefoil knot, a fundamental prime knot with three crossings, serves as the central focus of this presentation. We present our attempt at developing a criterion for determining when a knot is divisible by the trefoil, that is, when it can be decomposed into a connected sum involving the trefoil as a factor. Our approach integrates three main themes: (1) essential prerequisites in knot theory, (2) polynomial invariants such as the Conway and Alexander polynomials, and (3) applications of group theory, particularly the connection between knot divisibility and surjective homomorphisms of knot groups. We further examine a subset of knots called torus knots, presenting two theorems for the families $T(p, 2)$ and $T(p, 3)$ which we then generalize, with proofs leveraging polynomial algebra and roots of unity. Finally, we unify these ideas by exploring implications between tricolorability, knot polynomials, and surjective homomorphisms, drawing on foundational work by Crowell-Fox, Kitano-Suzuki, and others. Our results highlight relationships that characterize divisibility by the trefoil knot.

Rescuing polynomials: a differential topology perspective

Frédéric-Alexandre Lacasse

Topology is often presented as a sort of variant of geometry, except that, unlike geometry, topology is not concerned with local invariants: curvature and torsion are no longer of interest. Instead, topology focuses on properties preserved under "deformation", that is, up to homeomorphism : a continuous bijection, with a continuous inverse function.

Differential topology goes a step further by extending differential and integral calculus to more general structures than the usual Euclidean spaces $\mathbb{R}^n$. These structures are known as differentiable manifolds.

After introducing some basic concepts and tools from manifold theory, I will present a surprising application—remarkable both for its ingenuity and originality: a proof of a famous theorem about polynomials!

Geometric Tomography of Convex Polygons using Piecewise Analytic Hedgehogs

Brendan Matthews

The main idea of this talk is determining uniqueness of 2 convex polygons in $\mathbb{R}^2$ by utilizing non-central sections of piecewise analytic hedgehogs. Starting from the basic idea of convexity for bodies in $\mathbb{R}^2$, to then defining the support function and supporting line of a convex body. This leads into defining hedgehogs and a brief overview of a new result.

Double Centralizer Theorem for Central Simple Algebra

Jason Fang

If you were given a non-commutative ring R, and a subring S, we can define $C_R(S)$ to be the set of element in R that commute with all elements S. Formally, $C_R(S) := \{x \in R : sx = xs \text{ for all } s \in S\}$. It is not hard to deduce that $C_R(S)$ is a subring of R and $S \subseteq C_R(C_R(S))$. Is $S = C_R(C_R(S))$?

No! ehehehehe.

However, we will be discussing about a case where it is true, namely when we are working with (finite dimensional) central simple algebra and simple subalgebras. We will hopefully go over a rough proof, and, if there is still time left, talk about some applications.

Circle method on Waring Problem and Hasse Principle

Cuiwen (Frank) Zhu

The Hardy-Littlewood circle method is one of the most powerful tools in analytic number theory, originally developed to study additive problems such as Waring's Problem. In this talk, we present a sketch of how the circle method is used to estimate the number of representations of a natural number n as a sum of k-th powers. In the final part of the talk, we briefly discuss how the circle method connects to the Hasse principle, particularly in the context of local-global solubility of Diophantine equations.

Lie Theory and the Connection Between Analysis, Algebra, and Geometry

Jester Cabading

At a surface level, Lie theory studies Lie/Linear algebraic groups and Lie algebras. However, as I will show in this talk, this field of math is even more compelling because of how it connects various branches like geometry, algebra, and analysis together.

Perfect Numbers

Aria Duggan

A perfect number is a number that equals the sum of its proper divisors. Determining if all perfect numbers are even is an open problem. This presentation is about some cases of this problem depending on the amount of prime factors in a number. No background knowledge is needed.

The Classification of Finite Real Reflection Groups

Ena Bahk-Pi

We will explore the classification of finite real reflection groups, which are symmetry groups generated by reflections in Euclidean space. By using root systems and Coxeter graphs, we arrive at the complete classification into ten distinct types.

The isoperimetric problem in the plane

Axel Liu-Bai

Given a length, which curve encloses the greatest area? You probably learned in school that the answer is the circle, but why is that true? This is a classical question known as the isoperimetric problem (in $\mathbb{R}^2$). In this talk, we will explore the history of this problem, some of the flaws in the earlier "proofs"

and how they were fixed. Then, I will present a proof of the isoperimetric inequality. Finally, we will see how this inequality can be used to prove other results.

Colouring Knots with S4

Tsepella Goff-Martin

Fox n-colouring is used to find knot invariants. Using Wirtinger presentations we use permutations and conjugates to colour various knot types. The goal was to find a pattern for knot types. This method of colouring was used on torus knots. We found empirically that torus knots do not have non-trivial S4-colorings. This was employed on other knot families. We tested the capabilities of AI to help code a system. The automatic extraction of Wirtinger equations was tested on the trefoil, figure-eight, stevedore, and pretzel diagrams. The program output produced S4 colourings for the trefoil and stevedore knot, but no non-trivial solutions for the figure eight knot. This colouring of knot families was limited due to AI hallucinations. This part of a project done with James Qiu and Amos Zhang for the University of Toronto Math Mentorship Program.

The Undecidability of the Word Problem for Groups

Nathan Acheampong

In the 1930s, Alan Turing and Alonzo Church laid the foundations for modern (theoretical) computer science, and in particular, the theory of computation. It is not long before the famous halting problem, among other problems, are found to be undecidable, but these first are mostly from mathematical logic if not from the theory of algorithms. The earlier defined word problem for groups, along with the group isomorphism and conjugacy problems, is one of the first natural problems from a classical field of mathematics to be proven to be undecidable. P. Novikov and W. W. Boone first proved this result in the late 50s, which was later simplified by J. L. Britton in 1963. After reviewing the concept of decidability, we will explore the simplified proof by S. Aanderaa and D. E. Cohen (1980) which uses a different model of computation: modular machines.

Coloring Pretzel Knots Under The Symmetric Group Of Degree 4

Amos Zhang

Pretzel knots $P(p, q, r)$ form a well-known family obtained by joining three twisted strands with p, q, r half-twists each. A group-coloring of a knot labels each arc by an element of a group G so that at each crossing certain relations holds. Taking $G = S_4$ (the symmetric group on 4 letters) yields a richer invariant than the classical p-coloring. In fact, a pretzel knot $P(p, q, r)$ admits a nontrivial S_4-coloring

using only transposition labels exactly when two of p, q and r are multiples of 3 and a third is congruent to $0, 2, 3$, or 4 (mod 6). This research, completed under the Math Mentorship Program with Tsepella Goff-Martin and James Qiu, under the guidance of Malors Espinosa will outline the ways pretzel knots can be colored in such a manner, including S_4-coloring with other conjugacy classes.

Post-Quantum Cryptography: The Spine of the Supersingular Ell-Isogeny Graph

Taha Hedayat

Supersingular ell-isogeny graphs defined over algebraically closed finite fields are used in numerous cryptographic protocols considered to be quantum-resistant. Much of the security analysis of these schemes assumes the supersingular ell-isogeny graph to have a random structure. This is a questionable assumption which may have detrimental consequences if false. In this talk, we present the joint work with Dr. Sarah Arpin and Dr. Renate Scheidler that studies the subgraph known as the spine. We provide explicit conditions on the parameters of the supersingular ell-isogeny graph which result in detailed descriptions of the spine. As such, providing some insight into the graphical structure of the supersingular ell-isogeny graph. These findings challenge the assumption of the inherent randomness of the graph. We also discuss numerical data collected about the spine's position within the full supersingular ell-isogeny graph. In particular, we look at the number of vertices from the spine that lie in the center of the supersingular ell-isogeny graph. Interestingly, through its wave-like pattern, this data seems to show a certain level of randomness still present in the supersingular ell-isogeny graph.

An introduction to Gelfand-Kirilov dimension

Sara Nayar

A look at how we extend dimension from vector spaces to algebras, both commutative and noncommutative. We discuss finite vs finitely-generated algebras, and the notions of Krull dimension, and one of its noncommutative extensions, Gelfand-Kirilov dimension.

Analysis, Probability, and Statistics

LEAP: A Stochastic Microsimulation Model for Asthma Outcomes and Policy in Canada

Mark Ewert

LEAP (Lifetime Exposures and Asthma outcomes Projection) is a stochastic microsimulation model designed to evaluate asthma-related health outcomes and policies in Canada. It simulates individual life histories from 2001 onward, incorporating probabilistic transitions for asthma incidence, symptom control, exacerbation severity, and healthcare costs.

Each virtual individual is assigned attributes such as age, sex, and early-life exposures (e.g., antibiotics), and progresses annually through a set of probabilistically determined health states. Structural equations are parameterized using national datasets and literature, with model components including logistic, Poisson, and Dirichlet-multinomial regressions.

The model captures key interactions between risk factors and outcomes, and aligns closely with Canadian demographic and epidemiological data through internal validation and calibration. A recent reimplementation in Python has improved usability and extensibility.

Ongoing development introduces spatially resolved PM2.5 air pollution as a dynamic environmental exposure, enhancing LEAP's ability to quantify the impact of changing environmental conditions on asthma outcomes. This project illustrates how applied probability and simulation modelling can support public health decision-making and evaluate long-term population-level policy interventions.

Generalized Symplectic Vector Spaces and Operads

Kai Boisvert

A symplectic vector space consists of a vector space V and a nondegenerate 2-form on V. In this talk, we will generalize this notion to multisymplectic, polysymplectic, and finally vector-valued multisymplectic vector spaces. We will then look at operads, which are special algebraic structures consisting of collections of operations and their compositions. The goal is to highlight an operadic structure which naturally arises from the generalized symplectic vector space.

Identification of Interventional Distributions

Carol Wu

Causal inference seeks to uncover causal relationships between variables from observational data. However, this task is complicated by two fundamental challenges:

(1) unobserved confounding, where hidden variables create spurious correlations (correlation $\neq$ causation).

(2) equivalence classes of causal graphs, where distinct graphs imply identical observational distributions.

While randomized interventions can isolate direct causal effects, such experiments are often impractical or unethical—for example, we cannot force pregnant mothers to smoke to study its effects on their children. To address (1), mathematical tools like Pearl's backdoor/frontdoor criteria and Shpitser's covariate adjustment criterion sometimes allow us to derive interventional distributions *without* physical interventions, provided we assume a plausible causal structure. In this talk, we will derive and intuit these adjustment criteria, discuss their limitations, and explore potential extensions for broader application.

The Martingale Convergence Theorem - A Lesson on Gambling

Leo Guan

In this talk, we will present the Gambler's Ruin problem, and we will perform an analysis to show that in a fair game, the gambler will always ruin themselves. We will then generalize this problem into what is known as Doob's Martingale Convergence Theorem. Finally, we will see a real life application of this theorem.

Kernels, Hilbert Spaces and Hypothesis Testing

Aaron Wei

Positive semi-definite kernels, or simply kernels, are pairwise comparison functions which provide powerful techniques for tasks involving non-linearity (e.g. regression, smoothing, pattern analysis). In this talk we will use analyze the theoretical foundations for kernel methods, in particular each kernel induces a reproducing kernel Hilbert space (RKHS) which allows us to make connections with functional analysis. We will then show how operators and norms between RKHS are the building blocks for a variety hypothesis testing problems.

The Polynomial Method and the Finite Field Kakeya Problem

Madeline Forbes

The finite field Kakeya problem asks for the minimal size of a set in $\mathbb{F}_q^n$ containing a line in every direction, and was originally posed by Thomas Wolff as an analogue to the Euclidean Kakeya problem. It was conjectured that such sets have a cardinality bounded below by $c_n q^n$, where c_n depends only on n. In

2008, this conjecture was resolved by Zeev Dvir using what is now known as the polynomial method. In this talk, we will discuss the polynomial method and some of its applications, including Dvir's proof of the finite field Kakeya conjecture.

Glasses for the Mind's Eye

Nicholas Rees

How do you visualize something in four dimensions? Visualizing complex functions has often suffered from this problem, causing us to lose much of the intuition we have of what elementary functions look like when we change x to z. We'll explore a 2D visualization tool to aid us in understanding what these "complex beauties" look like, and how they are particularly helpful when studying analytic (meromorphic) functions. With a stroll through functions that both have mathematical significance and aesthetic beauty, we will see how this visualization helps us intuitively understand complex dynamics, algebraic symmetries in automorphic functions, and a deep result concerning Riemann zeta function just through pictures. If time permits, we'll see how our visualization techniques extend to functions with branches, that is on Riemann surfaces. Regardless if you're looking to expand your understanding or just want to look at pretty pictures for 20 minutes, anyone will be able to take something away from this talk (although a first course in complex variables certainly won't hurt)

The High-Low Method in Incidence Geometry

Matthew Bull-Weizel

Incidence geometry in it's simplest form is the study of the intersection of points and lines. This work can take place in a variety of settings, euclidean space, finite fields, local rings. However when these settings are "nice enough" the Fourier transform is well behaved, and we make use of the "High-Low Method".

The High-Low Method abuses the fact that if we have many lines parallel, then the Fourier transform of the indicators of those lines is supported on the dual tube passing through the origin. And so now we may study the incidences of these dual tubes in frequency space, treating cases where the frequency is high, and low separately.

This talk will give a short proof of an point-line incidence bound using the High-Low Method, and further dive into how the High-Low Method can be used in

$$\mathbb{Z}_p$$

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Solving Integer Programming Problems with Quantum Computing

Chunxin Zheng

Integer programming plays a key role in many areas of science and industry, including scheduling, logistics, and network design. However, these problems are often difficult to solve efficiently using classical methods. In the talk, I will explore how quantum computing can be incorporated for solving such optimization problems. The main idea is to transform an integer programming problem into a Quadratic Unconstrained Binary Optimization (QUBO) model, then be converted into an Ising model, which is compatible with the structure of many quantum computing devices, such as quantum annealers. I will introduce the mathematical steps involved in these conversions and provide simple examples to illustrate the process. If time's allowed, I'll also discuss the advantages and current limitations of using quantum hardware for solving these problems, offering a glimpse into how quantum computing could shape the future of optimization.

Unpacking The Kakeya Conjecture: Problems and Approaches

Sushrut Tadwalkar

The Kakeya conjecture is an infamous problem in Harmonic Analysis and Geometric Measure Theory, that has to do with estimating the size of a specific class of sets in $\mathbb{R}^n$. This conjecture was recently resolved in 3 dimensions: the journey to this has been a long one, spanning multiples decades, and has resulted in the creation of a lot of interesting mathematical machinery. In this talk we will discuss some of the milestone advancements made in the pursuit of solving this conjecture, and highlight the main mathematical strategies that were employed.. There are no pre-requisites, but familiarity with simple Euclidean geometry and some maximal function theory would enhance understanding.

Applied Mathematics, Mathematical Physics

Spectral problems related to solitary waves of the nonlinear Schrodinger equation and their stability

Mark Choi

The nonlinear Schrodinger equation (NLSE) models wave propagation in various physical systems, such as laser beams in Kerr media and Bose-Einstein condensates. A (ground state) solitary wave is a solution to the NLSE with a positive, smooth, and radially-symmetric shape that is preserved in time. To study the stability of these solutions, we perturb the initial condition in a way that introduces the linearized operators L, L+, and L-, whose spectral properties determine the linear stability of the wave. In this talk, I will introduce the NLSE, define key notions of stability, and describe a perturbed eigenvalue problem for L+ that governs the interaction between solitary waves and a distant external potential, and present some numerics for this problem.

A look at AIM with new applications

Mingyue Guo (Gloria)

The Asymptotic Iteration Method (AIM) is an iterative method for solving linear second-order differential equations both analytically and approximately. It has found extensive applications in theoretical and mathematical physics, particularly in the study of eigenvalue problems. In this work, we present a theorem that clarifies AIM's effectiveness for solving eigenvalue problems. By transforming a linear differential equation into normal form, we outline a method for generating analytical solutions for a wide range of quantum potentials. Particular attention is given to the generalised Hypergeometric Equation by providing a mathematical form for AIM sequences at an arbitrary n-iteration.

Spectral Clustering and Image Segmentation

Jiwei Peng

Any finite dataset can be transformed into an undirected, weighted simple graph by representing each data point as a vertex and assigning a weighted edge between every pair of vertices to reflect the level of similarity between them. By analyzing the matrix representation of this similarity graph through the lens of numerical linear algebra, one can determine an optimal partitioning of the graph that maximizes the similarity within subgraphs while ensuring the dissimilarity between them. This technique is known as spectral clustering. One practical application is image segmentation, where distinct objects within an image can be detected and extracted using this method.

3D Printing Manipulatives in Mathematical Education

Manuela Salerno and Niv Karo

3D printing and additive manufacturing have reshaped prototype development and enabled tailored solutions across industries—from healthcare to aerospace—and now offer powerful opportunities in education by making learn- ing interactive and hands-on.

Through our work, we have learned to utilize 3D printing software to produce manipulatives that help illustrate key concepts that appear throughout McMaster's undergraduate calculus sequence. We have also designed specialized lesson plans surrounding these 3D-printed models. Our goal is to introduce a more active, inquiry-oriented learning approach to the classroom, which we hope to achieve by allowing students to work in groups with the manipulatives on more ill-defined problems that they may not have encountered before. These lesson plans, which were formulated through backwards design, will promote a much more interactive, hands-on engagement with course concepts, where the students will be given more autonomy over their work and learn through problem solving. We hope to illustrate why 3D models can be beneficial to students and give insight on how they are designed, developed, and printed.

Optimization and Monotone Operator Theory

Manuel (Zicheng) Wang

Many modern optimization algorithms—from simple alternating projections to accelerated methods like FISTA—can be re-interpreted through the lens of monotone operator theory. In this talk we give a guided tour of that connection. We will see how projection and proximal operators arise as resolvents of maximally monotone operators, why Minty's theorem guarantees their well-posedness, and how forward-backward splitting unifies ISTA, FISTA, and related schemes. Along the way we highlight geometric intuition, classical convergence results, and a few open questions that remain surprisingly subtle. The talk is self-contained and aimed at undergraduates who have seen basic linear algebra and real analysis.

Cells Navigating in Complex Environments: Models and Simulations

Edwin Huras

Cells are remarkably effective navigators, moving through crowded and complex environments during processes like embryonic development, immunity, and cancer spread. Despite their size and apparent simplicity, different cells employ a surprising range of motility strategies, from crawling and flagellar swimming to sensing and shaping chemical gradients that guide their paths. This talk will introduce key mechanisms of cell movement and show how they can be described using partial differential equations and simulated in different computational frameworks. By combining experimental videos with simulations,

we will explore how cells find their way through mazes, avoid dead ends, and form dynamic clusters. Finally, we will discuss how these ideas are being applied to broader topics such as modelling human crowd dynamics.

Reservoir Computing: It Just Works!

Michael Alexander

In this talk I will discuss reservoir computing, a strange computational framework where one uses an unpredictable, high-dimensional dynamical system to make predictions on a smaller one. Reservoir computing has long been a favorite of computing on atypical hardware, and I'll survey some current efforts such as quantum systems. Finally, I'll give some future directions for reservoir computing and maybe talk about some active research, time permitting.

Fractal Geometries and Architecture

Chris Richard

To better characterize and study natural geometries, the Polish born French American mathematician Benoit Mandelbrot proposed a new field of geometry on fractal shapes; these shapes do include lightning bolts, trees, and seashells. In Mandelbrot's book The Fractal Geometry of Nature, he makes frequent reference to the Eiffel Tower and other buildings designed by the Beaux Arts school of architecture asserting that these manmade structures, too, have fractal geometries in their design.

Methods used in the analysis of fractal geometries are useful to architects working to find quantitative ways to represent the style of various structures. This is results in a more attractive and natural design. Modern architects use computer programs to measure the degree of self similarity of buildings and their relation to metallic means (such as the golden ratio) which have long been known to be aesthetically pleasing.

The aim of this presentation is to introduce the concept of fractal geometries and highlight their presence in both nature and architecture. MatLab code will be used to analyze the dimension of non-Euclidean shapes as an example for how fractal dimensions are calculated. In conclusion, the beauty of the connection between fractal geometries, quantitative linguistics, metallic means, and architecture will have been brought to light in a visual and informative manner.

Mathematical Modeling and Reinforcement Learning for Strategic Ocean Plastic Retrieval

Maria Chzhen

This work explores the application of mathematical modeling and Reinforcement Learning (RL) to optimize the strategic collection of ocean plastic by autonomous agents. We formalize the problem of navigating a simulated ocean environment with varying plastic densities as a Markov Decision Process (MDP), defining key components including the state space representation of the agent and plastic distribution, the agent's action space, and a carefully designed reward function that incentivizes efficient plastic retrieval while accounting for operational costs. Using the Proximal Policy Optimization (PPO) algorithm, we train a policy for a single autonomous collector within this simulation. Preliminary results demonstrate the agent's ability to learn non-trivial collection strategies. This project highlights the power of combining mathematical abstraction and modern RL techniques to address pressing environmental challenges and lays groundwork for more complex simulations involving dynamic environments and multiple agents.

Understanding Opioid Mortality during COVID-19 - An Individual-Based Model

Carter Clifton

The opioid epidemic is a global public health crisis, with significant impacts across Canada, including over 50 thousand opioid-related deaths since 2016. During the COVID-19 pandemic, the crisis deepened: in Toronto, the number of opioid related fatalities in Toronto nearly doubled from 2019 to 2020, and has remained high since.

Here we describe an individual-based dynamic model of the population of people who have opioid use disorder to determine overdose and overdose fatality, depending on each individual's demography, use history, and use of harm reduction services. The model is first parametrized to urban Toronto in 2019. Then the factors known to have changed during the pandemic are implemented in the model to quantify changes in mortality, to explain the increase in overall overdose fatality.

We find that changes to harm reduction services do not account for the majority of the increase in overdose fatality. Instead, we explain the increase due to a simultaneous increase in the presence of fentanyl in the drug supply. We use available data from urban Toronto to predict the amount of fentanyl in the supply in 2019 versus the 2020-2023 period (during and post-pandemic), and the fatality risk factor associated with its presence.

Can AI Teach Proof? Student experiences with a custom chatbot in an introductory Algebra course

Taha Hedayat

As artificial intelligence (AI) becomes increasingly integrated into education, it is imperative to assess the effectiveness of AI tools in supporting student learning. This presentation details a multi-stage research study designed to evaluate the efficacy of AI tools for students learning proof-based mathematics in an introductory abstract algebra course (MATH 315). The study commenced with a preliminary exploration phase where two students familiar with the course material assessed the strengths and weaknesses of various AI chatbots in assisting with learning. The findings of this phase informed the development of a specially prompted AI chatbot, called the Intelligent Tutoring System (ITS). Part 1 of the main study consisted of MATH 315 students regularly using the ITS to study course content, then were asked to complete two surveys evaluating their experience. Following part 1 of the study, 13 and 15 participants completed surveys 1 and 2, respectively. Most participants found the tool helpful in their study routine, and said that they would recommend it to a student taking the class in the future. Part 2 of the study consisted of 15 students studying for a mock exam on a new topic, either using the ITS or traditional study methods. The mock exam was followed by a focus group session where students discussed their experience using the ITS in comparison to studying traditionally. Preliminary results from the focus group session will be presented.

Combinatorics, Graph Theory, and Logic

Automating Enumeration: Chord Diagrams from Combinatorics to Code

Stephanie Penner

Enumerative combinatorics works with objects like binary strings or permutations to ask questions like, "How many objects of size n are there?" and "What does the counting sequence look like as the size of the object grows arbitrarily large?". These types of questions can be analyzed using algebraic tools, like generating functions. To use these tools, we often need a way of decomposing the class of objects into simpler ones that have more known information. This can be tricky, but a recently developed framework called "Combinatorial Exploration" aims to automate this process. In this talk, I will discuss how this process works, and how I'm adapting it to apply to chord diagrams.

Eternal Domination of Graphs

Ewan MacDuff

In my talk, I will briefly introduce graph theory to anyone who is unfamiliar, and then explain the idea behind domination number and eternal domination. I will then explain the upper and lower bounds on eternal domination number, and give examples of graph classes which force those bounds to be tight.

Hermitian adjacency matrix

Jonny Aloni

Joint work with Saieed Akbari, Maxwell Levit, Bojan Mohar, and Steven Xia. The Hermitian adjacency matrix is a way to transform a digraph into a matrix, properties of the graph are closely related to its matrix. The way to construct the Hermitian adjacency matrix is by having a digraph D and a complex unit ω applied on the following rule

$$H_\omega(D)_{u,v} = \begin{cases} \omega & u \to v \\ \overline{\omega} & u \leftarrow v \\ 0 & u \not\sim v \end{cases}$$

In general we pick $\omega = \frac{1+\sqrt{3}i}{2}$ a 6th root of unity, because $\omega + \overline{\omega} = 1$, which adheres to models of mixed graph where two arcs in opposite direction are equivalent to an edge.

This talk will introduce the Hermitian adjacency matrix and its properties. As well as present results on non-bipartite digraphs with symmetric spectrum, and digraphs with 2 distinct eigenvalues.

Kemeny's Constant and Braess' Paradox for Graphs

Emma deBlieck

Can adding what seems like a shortcut slow down an entire network? In this talk, we will explore Kemeny's constant, a measure of the average length of a random walk on a graph. Several ways to compute it will be presented with examples, and we will investigate how Kemeny's constant changes when a new edge is added to a graph. Surprisingly, adding an edge does not always improve connectivity, revealing a counterintuitive phenomenon known as Braess' paradox.

Perfect State Transfer in Symmetric Square of the n-cube

Hanyang Xiao

Let G be a (V, E) graph. Then, the kth-symmetric power of graph G is denoted as $G^{\{k\}}$. $G^{\{k\}}$ is the graph with vertex set of all k-subsets of $V(G)$, where two vertices are adjacent in $G^{\{k\}}$ whenever their symmetric difference is in $E(G)$. Usually, we call $G^{\{2\}}$ as the symmetric square of G. Also, we denote the n-cube as Q_n. In this presentation, we will talk about the characteristics of the perfect state transfer to check if the symmetric square of n-cube admits perfect state transfer. We achieved a proof for n=2 as an example and two disproves for n=3,4 by a computer program. Further, there are some interesting results by observation.

Alice in Quadraticspanningforestidentityspace

Kai Choi

Quantum field theorists study Feynman periods, which are obtained by integrating expressions related to the spanning tree polynomials of graphs known as Feynman diagrams. Without the ability to compute Feynman periods, it would be virtually impossible for high-energy physics groups like CERN to collect meaningful data from their particle accelerators.

Much of this is beyond the concern of this talk, as my will to do physics died painfully in a Zoom classroom during the pandemic. Fortunately, deep thoughts about quantum field theory often lead to combinatorial play, and Feynman periods live at the door to a network of questions about set partitions, determinantal identities, spaces of polynomials, the all-minors matrix-tree theorem, Lewis Carroll, and the interplay between them. Many of these questions are unanswered, but often they are accessible to undergraduates like you and me. I will present these questions and their relevant background in the context of work I'm doing this summer on spaces of quadratic spanning forest identities, supervised by Dr. Karen Yeats. If time permits, I will share the current progress of the work. No physics background required!

Large numbers and proof theory

Edmond Yu

Large finite numbers commonly seen in math usually are of the form of a power or double exponent. Larger numbers (for example, those that require multiply iterated exponents) are generally not too common. For a while, Grahams number was cited to be the largest (natural) number to appear in a math problem.

Here we present a way to create even more ridiculously large numbers. To do so we require some set theory and use ordinals to associate with how fast a function grows.

As we discover larger and larger numbers and how we can compare them, we approach the limit of provability of certain theories and run into ordinal analysis. This drives us to use even more complicated set theory and slowly becomes a philosophical question of "what is real".

2-dimensional partial orders and ultrahomogeneity

Mithuna Threz, Arul Milton

The set of rational numbers, when viewed as a linearly ordered structure, is characterized by its high degree of symmetry – it is "ultrahomogeneous". We consider two-dimensional versions of this fact, and show that the structure $(D_2, <, <_1, <_2)$, where D_2 is a dense subset of the rational plane without colinear points, $<$ the product partial order, and $<_1$, $<_2$ two linear orders that realize the product order, is also ultrahomogeneous.

Introduction to Interdiction Optimization

Amir Dadpour

Bilevel Optimization is a generalization of optimization models with some game theory flavour. In Bilevel Optimization, two players are optimizing over the same universal domain such that their choices affect each other's feasible regions. Interdiction optimization is special case of Bilevel Optimization with the two players having orthogonal objectives (i.e. they are trying to hurt each other as much as possible) In this talk, I will go over the basics and definitions of Bilevel and Interdiction Optimization along with some established and new methods to solve them. No promise on the solvers to be polytime though!

Enumerating Permutations in Even and Odd Symmetric Groups

Yuewen Luo

Let $\alpha(n)$ denote the number of perfect square permutations in the symmetric group S_n. The conjecture $\alpha(2n+1) = (2n+1)\alpha(2n)$, provided by Professor Richard P. Stanley, was proved by Joseph Blum using a generating function. In this talk, we present an alternative and combinatorial proof for this conjecture, which reveals structural insights about permutations and their disjoint cycles. At the same time, we demonstrate that all permutations with an even number of even cycles in both S_{2n} and S_{2n+1} can be categorized into three disjoint types that correspond to each other.